Implementation of the Safe and Sustainable by Design concept for industrial impact

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Safe-and-Sustainable-by-design

Working description from the Chemicals Strategy for Sustainability:

At this stage, **safe-and-sustainable-by-design** can be defined as a **pre-market** approach to chemicals that focuses on providing a function (or service), while **avoiding** volumes and chemical properties that may be **harmful to human health or the environment**, in particular groups of chemicals likely to be (eco) toxic, persistent, bio-accumulative or mobile.

Overall **sustainability** should be ensured by **minimising** the **environmental footprint** of chemicals in particular on **climate change**, **resource use**, **ecosystems** and **biodiversity** from a lifecycle perspective.



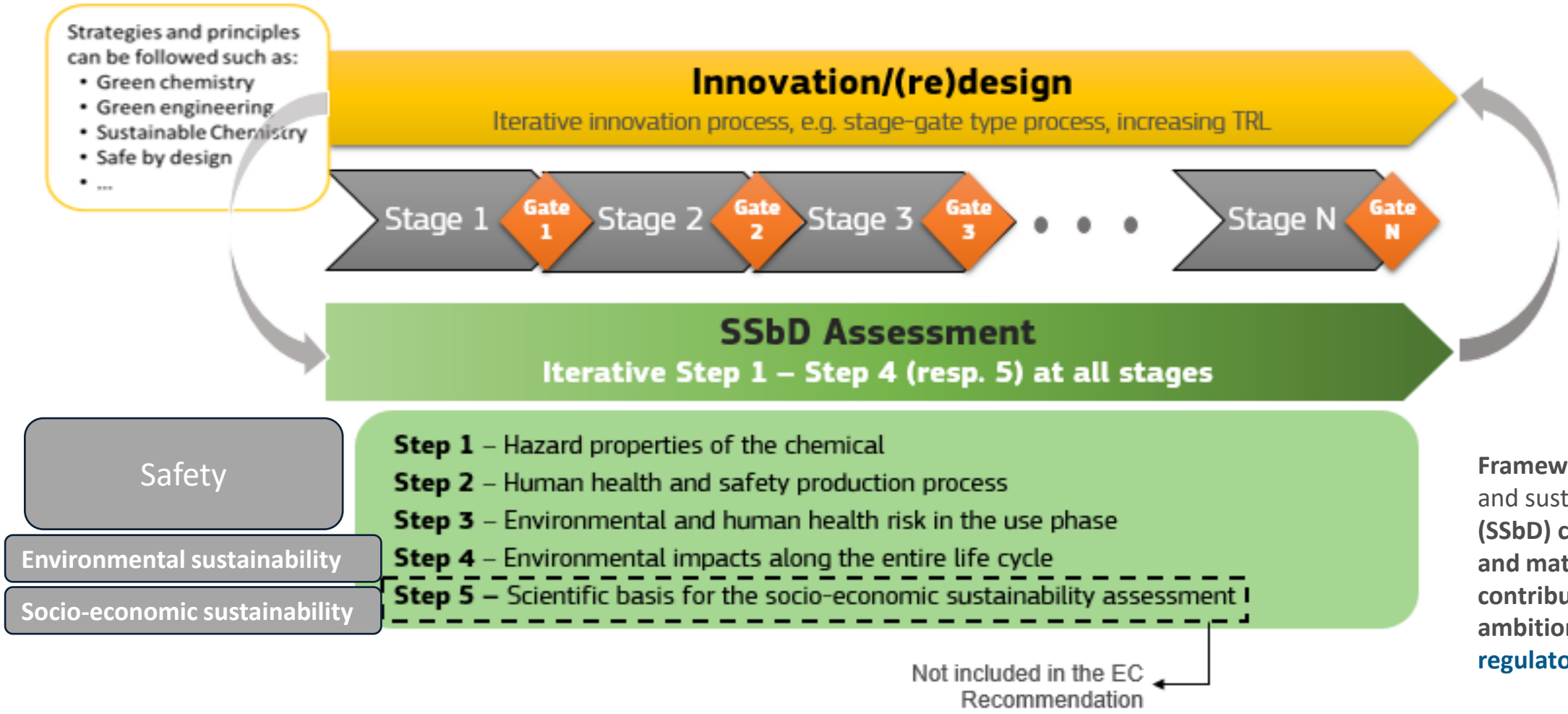
A system approach integrating

- Safety
- Climate neutrality
- Circularity and functionality of materials, products and processes throughout their lifecycle

Safe and Sustainable by Design Chemicals and Materials. Revised framework (2025)

By integrating safety and sustainability from the earliest stages of innovation, the SSbD framework can support a more resilient, competitive and innovation based, future-proof industrial ecosystems.

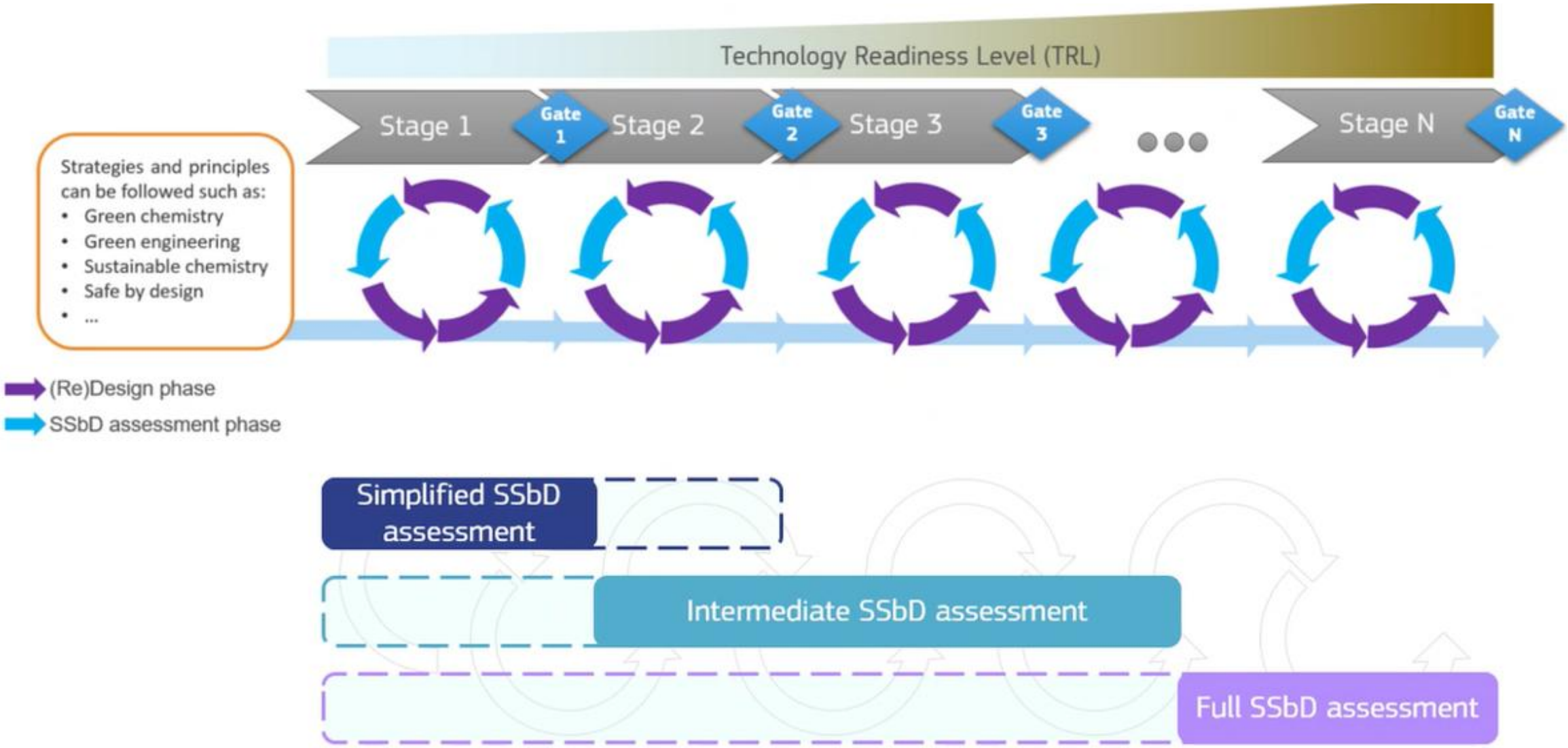
EC-JRC SSbD framework



Framework to define safe and sustainable by design **(SSbD) criteria for chemicals and materials** that should **contribute to achieve the CSS ambitions**, **beyond current regulatory compliance**.

European Commission, Joint Research Centre, Abbate, E., Garmendia Aguirre, I., Bracalente, G., Mancini, L., Tosches, D., Rasmussen, K., Bennett, M.J., Rauscher, H. and Sala, S., Safe and Sustainable by Design chemicals and materials - Methodological Guidance, Publications Office of the European Union, Luxembourg, 2024, https://data.europa.eu/doi/10.2760/28450, JRC138035

Iterative and tiered assessment



European Commission, Joint Research Centre, Abbate, E., Garmendia Aguirre, I., Bracalente, G., Mancini, L., Tosches, D., Rasmussen, K., Bennett, M.J., Rauscher, H. and Sala, S., Safe and Sustainable by Design chemicals and materials - Methodological Guidance, Publications Office of the European Union, Luxembourg, 2024, https://data.europa.eu/doi/10.2760/28450, JRC138035

Synergize and connect











Key outcomes from the value chains

- Valuable insights about SSbD's role in the different value chains and what needs to be addressed
- Case studies presented, introducing challenges and solutions/alternatives to each value chain
- A good understanding on the skills and knowledge already represented in the industry, in each value chain
- Gap analysis (including criteria): still lacking knowledge for the industry to fully embrace SSbD

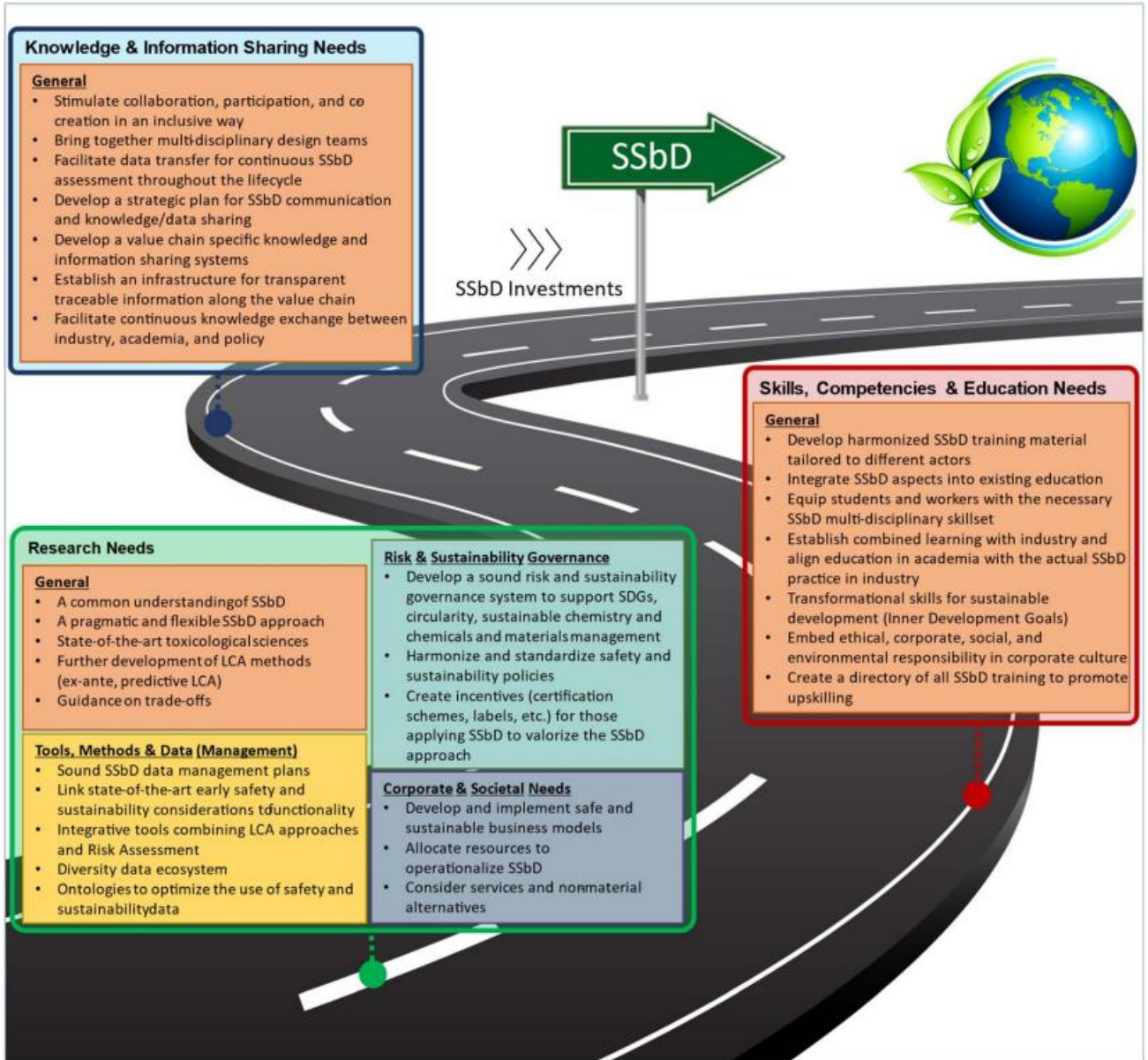


Gaps to address

- Lack of harmonized and easy to apply **tools and protocols**
- Unavailability of **open databases** with rich datasets
- Uncertainty of the safety impacts at **early stage** of the development
- Limited number of **internal experts**
- Problematic assessment for **complex products** with multiple materials and chemicals
- Full assessment at **late stages of innovation**



SSbD Supportive Roadmap





RSC Sustainability, 2024,**2**, 2833-2838

Research needs

- How to **implement** SSbD in the innovation process
- Further development of **LCA methods**
- **Updated/new** datasets and databases
- Complementary methods for **Risk Assessment** and **Sustainability Assessment**
- **Traceability** throughout the value chain (including communication to consumers)
- Implementation of **circularity** along the value chains
- Transition from conventional materials to **biobased alternatives**
- Research on **alternatives to the critical raw materials**
- Development of efficient **recycling methods** and technology
- Further development of **services** and **non-material alternatives**









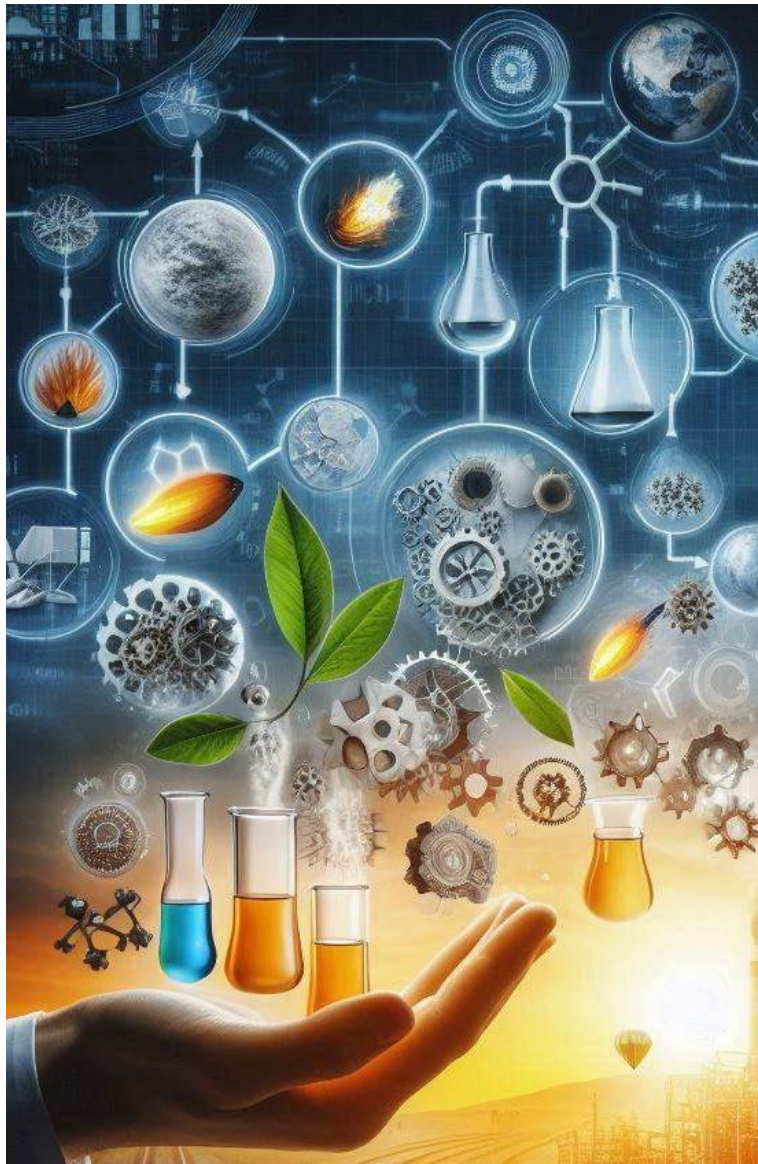
Skills and knowledge

- **Strong skills on safety** – different focuses per value chain – in-house/suppliers
- **Innovation/technical skills**: functionality/cost + sustainability
- **Skills on sustainability are less mature**, need for more internal knowledge, heavily depending on strategy/market demand/Investment capacity
- SSbD requires a **general understanding** of chemistry and material science and **system thinking**
- Knowledge and skills needed **along the value chain**
- Call for a **common tool** adjusted to the sector



Creating solutions

- Strong need for **cooperation** between industry, policy and science/research
- **Transfer knowledge** and best practice from larger organisations to SMEs
- **Cross-exchange of knowledge** between different value chains
- **Training** on methodologies, developing skills
- **Educational programs** at universities
- Educate the **general public**
- Consultancies
- Discussion on **trade-offs**
- Measures **against greenwash**



IRISS International SSbD Community

Knowledge and tools

- Learn from leading experts through webinars, workshops, guidelines, and practical tools to build capacity and support learning at different levels of expertise.
- Stay informed about the latest scientific findings, policy developments, and emerging trends in SSbD.

Community and networking

- Get access to virtual collaboration areas to connect with peers across industry, academia, and policy, and exchange insights, experiences, and solutions on specific topics or projects related to SSbD.
- Receive information about ongoing activities, events, and development within the community, and participate in forums that shape the future of SSbD.

Expert support

- Get guidance from experts in SSbD to support your implementation journey.
- Receive support in building consortia for research and innovation projects, finding suitable partners, and developing strong proposals.



500+ members are already part of the IRISS International SSbD Community!



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